

LITERATURE REVIEW

[1]. M. Rohini, G. Vijayvenkatajalapathi, S. Selvakumar, M. Sugavanesh, S. Viswanath, and R. Gowthamani, "Campus Sync - One Central Hub for Attendance, Events, and On-Duty Management," 2024 Ninth International Conference on Science Technology Engineering and Mathematics (ICONSTEM), 2024.

The proposed system, Campus Sync, introduces an integrated platform to manage attendance, events, and on-duty assignments in college environments. It features three core modules: Attendance Management, which enables multi-device attendance tracking with hourly configurations and statistical insights; Event Management, which simplifies scheduling and room allocation for campus activities; and On-Duty Management, which streamlines request handling and approval workflows. Developed using Flutter, the application supports multiple platforms and enhances accessibility. By unifying administrative functions in a single application, the system reduces paperwork, improves departmental coordination, and promotes efficient and streamlined campus management.

[2]. J. M. E. Paz, A. M. T. Maala, J. A. G. Panti, V. E. I. Hermoso, C. L. M. Melchor, and K. G. Kikuchi, "E-Ganapp: A Seamless Event Management System for Mapúa Malayan Colleges Laguna," 2024 22nd International Conference on ICT and Knowledge Engineering (ICT&KE), 2024.

This study presents E-Ganapp, a customizable and paperless event management system developed to streamline the event organization process at Mapúa Malayan Colleges Laguna. The system supports end-to-end event handling from online registration and digital evaluations to automated certificate issuance with minimal human intervention. Designed to enhance sustainability by reducing paper usage, E-Ganapp addresses common logistical challenges faced by institutional organizers. Testing results confirm its effectiveness in improving usability, efficiency, and performance, making it a reliable and eco-conscious tool for managing campus events.

[3]. B. Kamala, N. Swaminathan, G. M. Sanjay, and D. Jane Salome, "Event Horizon: Revolutionizing University Event Management with PWA," 2025 International Conference on Computing and Communication Technologies (ICCCT), 2025.

The Event Horizon (EH) system is a Progressive Web Application (PWA) designed to transform university event management by offering a role-based architecture with three user tiers: students, sub-admins, and super-admins. The platform includes a centralized events page, real-time chat for student collaboration, announcements for targeted groups, and comprehensive oversight tools for super-admins. EH also allows students to join interest-based groups and employs machine learning models for advanced report generation. By integrating communication, coordination, and data-driven insights into a single application, EH effectively addresses the limitations of traditional campus event systems and enhances overall engagement and operational efficiency.

[4]. R. Deepika, M. Selvam, and S. Gurumoorthy, "College Communication App - Chatster," 2023 Intelligent Computing and Control for Engineering and Business Systems (ICCEBS), 2023.

The study introduces Chatster, a mobile application designed to enhance communication within educational institutions. Addressing the need for streamlined and smartphone-friendly information sharing, the app enables students, parents, and staff to stay informed about exams, lectures, placements, events, transportation, and notifications. Developed using technologies such as Java, Android, and GPS, Chatster simplifies complex communication channels and boosts student engagement by offering event reminders and real-time updates. By promoting user-friendly access to institutional services, the application aims to reinvigorate student interest and participation in academic life.

[5]. Akansha Pansare,Athang Patil,Nikita Patil,Yatin Patil,Mrs. Aparna Bhonde "Smart College Event Management System Using MERN Stack "Mar 2023-Volume 11 Issue III International Journal for Research in Applied Science & Engineering Technology (IJRASET)

Traditional methods of managing college events using spreadsheets and manual records are inefficient and error-prone as events grow in number. Recent studies suggest that web-based event management systems can automate tasks like registrations, scheduling, announcements, and report generation. Based on these findings, the proposed Smart Event Management System aims to replace manual processes with an automated platform for efficient event handling.

[6]. S. Dhanasekaran; K. Shabareesh kumar; S. Vignesh; S. Nambivelnathan; R. Paulsudhan; Senthil Pandi S "An Intelligent Agent Based University Event Management System Using Harmony Search Strategy" 2024 Second International Conference on Advances in Information Technology (ICAIT)

Traditional event management in colleges has relied heavily on physical records, making it difficult to access specific information about events, participation, and achievements. Such manual systems are inefficient and prone to errors. Recent studies highlight the use of intelligent agent systems and algorithms such as the Harmony Search Algorithm to improve event planning and coordination. These systems enable centralized management of college activities including fests, workshops, and seminars, while providing features such as automated registration, attendance tracking, notifications, venue and resource allocation, online ticketing, and post-event feedback. Research shows that these online platforms improve communication between organizers and participants, enhance data security, and streamline the entire event management process. Building on these findings, intelligent systems are being developed to make event organization more efficient and user-friendly.

[7]. Nandakishore A., CampusConnect: A Digital Solution for Campus Event Management, Department of Computer Science, Bharata Mata College (Autonomous), Thrikkakara, Kochi-21, Kerala, India, March 2025.

Manual event planning in colleges involves paperwork, delays, and poor communication. Digital platforms like **CampusConnect** automate venue booking, approvals, cancellations, and notifications using verified institutional IDs. These systems provide real-time status, transparency, and faster processing, improving student engagement and optimizing campus resources by replacing inefficient paper-based processes.

[8]. Prof. Shriram Kulkarni , Aniket N Shinde , Aniket U Shinde , Mukteshwar Samdade , Pratiksha Pansare"College Resources and Event Management & Analysis "International Open-Access, Double-Blind, Peer-Reviewed, Refereed, Multidisciplinary Online Journal Volume 3, Issue 8,International Journal of Advanced Research in Science, Communication and Technology (IJARSCT)

Managing academic resources and events in colleges through traditional methods often leads to inefficiency and lack of accessibility. Recent studies emphasize the importance of integrated platforms that combine event management with academic services such as study

content sharing, project tracking, internship references, and alumni interaction. Research shows that such systems enhance communication, prevent duplication of work, and enable data-driven decision-making. Platforms built with modern technologies like Flutter and microservice architectures also ensure cross-platform support and optimized resource usage, making them more efficient and scalable.

Summary of Literature Review

S.NO	TITLE	METHODOLOGY	PROS	CONS
1	M. Rohini, G. Vijayvenkatajalapathi, S. Selvakumar, M. Sugavanesh, S. Viswanath, and R. Gowthamani, "Campus Sync - One Central Hub for Attendance, Events, and On-Duty Management," 2024 Ninth International Conference on Science Technology Engineering and Mathematics (ICONSTEM), 2024	Flutter-based integrated platform with three core modules: Attendance, Event, and On-Duty Management. Supports multi-device access and statistical insights.	Reduces paperwork; Cross-platform accessibility; Enhances coordination; Unified system for admin tasks.	Complex to implement and maintain; Initial learning curve.
2	J. M. E. Paz, A. M. T. Maala, J. A. G. Panti, V. E. I. Hermoso, C. L. M. Melchor, and K. G. Kikuchi, "E-Ganapp: A Seamless Event Management System for Mapúa Malayan Colleges Laguna," 2024 22nd International Conference on ICT and Knowledge Engineering (ICT&KE), 2024.	Paperless and customizable end-to-end event system including registration, evaluations, and automated certificate generation.	Eco-conscious; Efficient; Reduces human intervention; Improves usability and logistics.	Limited to institution-specific scenarios; Needs proper system testing and deployment.
3	B. Kamala, N. Swaminathan, G. M. Sanjay, and D. Jane Salome, "Event Horizon: Revolutionizing University	Progressive Web App with role-based access (student, sub-admin,	Centralized event page; Enhanced engagement; Intelligent	Resource-intensive; Requires ML model training and admin management

	Event Management with PWA," 2025 International Conference on Computing and Communication Technologies (ICCCT), 2025.	super-admin), real-time chat, group announcements, and ML-based reports	insights; User-specific access levels	
4	R. Deepika, M. Selvam, and S. Gurumoorthy, "College Communication App - Chatster," 2023 Intelligent Computing and Control for Engineering and Business Systems (ICCEBS), 2023.	Mobile application developed using Java, Android, and GPS to share real-time info across stakeholders.	Boosts student engagement; Real-time updates; Streamlines communication	Lacks complete event management features; Functionality limited to communication and alerts
5	Akansha Pansare,Athang Patil,Nikita Patil,Yatin Patil,Mrs. Aparna Bhonde "Smart College Event Management System Using MERN Stack "Mar 2023-Volume 11 Issue III International Journal for Research in Applied Science & Engineering Technology (IJRASET)	Web-based system built with MERN Stack to automate registrations, scheduling, announcements, and reporting.	Modern tech stack; Reduces manual errors; Real-time event handling	Needs backend/server setup; Technical expertise required
6	S. Dhanasekaran; K. Shabareesh kumar; S. Vignesh; S. Nambivelnathan; R. Paulsudhan; Senthil Pandi S "An Intelligent Agent Based University Event Management System Using Harmony Search Strategy" 2024 Second International Conference on Advances in Information Technology (ICAIT)	Centralized system using Harmony Search Algorithm to manage registration, attendance, notifications, ticketing, and feedback	Intelligent planning; Enhanced communication; Improves coordination and resource allocation	High algorithm complexity; Requires user training and digital readiness
7	Nandakishore A., CampusConnect: A Digital Solution for Campus Event Management, Department of Computer Science,	Verified institutional ID-based platform to manage bookings,	Transparency; Reduces paperwork; Accelerates approvals;	Requires integration with existing campus infrastructure; May face

	Bharata Mata College (Autonomous), Thrikkakara, Kochi-21, Kerala, India, March 2025.	approvals, and notifications with real-time tracking.	Improves resource usage	institutional resistance
8	Prof. Shriram Kulkarni , Aniket N Shinde , Aniket U Shinde , Mukteshwar Samdade , Pratiksha Pansare”College Resources and Event Management & Analysis ”International Open-Access, Double-Blind, Peer-Reviewed, Refereed, Multidisciplinary Online Journal Volume 3, Issue 8,International Journal of Advanced Research in Science, Communication and Technology (IJARSCT)	Flutter + Microservices-based integrated platform for events and academic resource management (projects, alumni, internships).	Combines academic and event modules; Scalable; Supports cross-platform deployment	High implementation complexity; Demands technical maintenance and planning